
SATELLITE MONITORING OF THE STATE OF ECOSYSTEMS IN ARCTIC REGION

Analysis of the Landscape Structure in the Tundra Zone for Western Siberia Based on Satellite Data

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Abstract—The landscape structure are investigated for typical and southern tundra subzones of western Siberia based on satellite images from Landsat and WorldView. The microlandscape types for the considered regions are selected and areas of microlandscapes are defined on the basis of satellite information and accounting data of field studies. The areas of the main wetland landscapes in the different types of bogs in northwestern Siberia are defined.

Keywords: tundra zone, western Siberia, landscape elements, satellite images

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INTRODUCTION

Earth remote sensing data are one of the sources of information on the state of terrestrial ecosystems. Space monitoring makes it possible to study ecosystems in regions, in which large-scale field measurements are difficult if not impossible to perform. The Arctic tundra zone is not an easily accessible region for ground-based measurements. Satellite data allow one to analyze the state of terrestrial ecosystems in this region (Elsakov, 2013; Kornienko and Yakubson, 2011; Reynolds et al., 2008), to study the dynamics of its vegetation cover (Tutubalina, 2006; Varlamova and Solov'ev, 2012; Olthof and Pouliot, 2010), to study cryogenic variations in the relief (Bryksina and Kirpotin, 2012; Kravtsova and Bystrova, 2009; Riordan et al., 2006), and to monitor both temperature and humidity characteristics of the atmosphere and the underlying surface (Pokrovkii, 2007; Golubyatnikov and Denisenko, 2006; Polyakov et al., 2009).

Current global climate change is most pronounced in both arctic and middle latitudes (Bindoff et al., 2013). Since the late 1980s, the rate of increase in the yearly mean temperature of the atmosphere over northern Eurasia has amounted to 0.4–0.5°C/10 yr (*Metody*, 2012). Over the last three decades, the yearly mean temperature of the atmosphere over northern Russia has increased by 0.8–1.3°C; moreover, the increasing air-temperature trend is 1.2–1.5 times larger for a long cold period in this region than that for a short warm one (Pavlov, 2008). According to estimates obtained using atmosphere–ocean general circulation models, significant climate changes (first and

foremost, an increase in air temperature and precipitation amount) are also expected to occur in the arctic regions (Collinz et al., 2013).

Carbon stocks in the soil organic matters in the high latitudes of the Northern Hemisphere are estimated at 1400–1850 Gt (McGuire et al., 2009). Climate warming in this region will result in accelerated degradation processes in the permafrost soil of this region and will significantly affect the intensity of the fluxes of carbon exchange between the atmosphere and tundra ecosystems. In particular, as a result of an increase in the depth of seasonal thawing of the Arctic soil, carbon conserved in the upper layers of permafrost soil will be included in an active biochemical cycle. Expected climate warming may result in a significant increase in both carbon and methane emitted to the atmosphere from the ecosystems of this region.

The emission of greenhouse gases from wetland ecosystems in the northern region depends significantly on a combination of microlandscapes that form every ecosystem. Generalized experimental estimates of both methane and carbon emissions from the tundra ecosystems of the high latitudes of the Northern Hemisphere (Kazantsev, 2013; Naumov, 2009; Karelin and Zamolodchikov, 2008) imply a significant difference in the intensities of these gas fluxes into the atmosphere from different microlandscapes of the ecosystems under consideration (for example, the emissions of methane from frozen hillocks and swamplands may differ by an order of magnitude). In order to correctly determine the regional emission of greenhouse gases from northern ecosystems according to experimental pointwise data, it is necessary to know

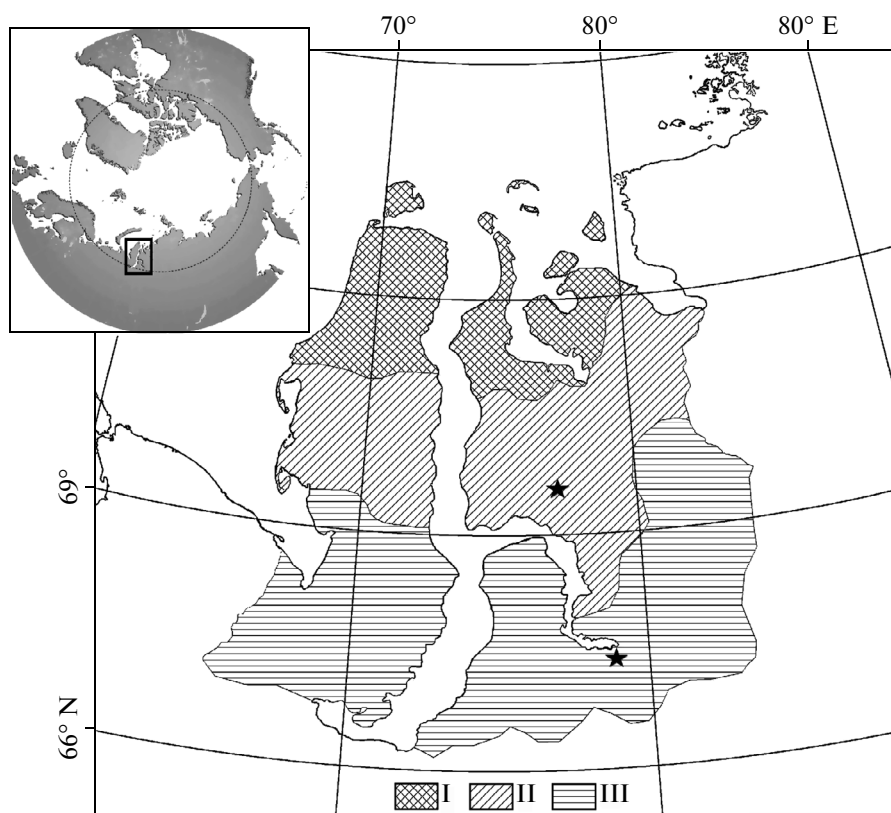


Fig. 1. Bioclimatic zones of tundra in western Siberia and location of the key field observation sites. Tundra subzones: (I) arctic tundra, (II) typical tundra, and (III) southern tundra. The observation sites are marked with asterisks.

not only the areas of different wetlands and mineral soil ecosystems of the region under study, but also the area ratios of their microlandscapes.

In this work, the landscape structure of the tundra ecosystems of western Siberia has been analyzed on the basis of data obtained from spectrozonal satellite images and field measurements.

OBJECTS AND METHODS OF STUDYING

According to the bioclimatic zonality of western Siberia (Liss et al., 2001), the tundra zone of this

region is subdivided into arctic, typical, and southern subzones (Fig. 1). The territory under consideration is characterized by a cold humid climate: annually mean air temperatures are negative and the amount of annual precipitation is about 300 mm (Table 1). The tundra ecosystems of western Siberia occupy over 340 000 km². Flat grounds with polygonal and flat-palsa bogs are characteristic of this region (Table 2). The relief of wetlands is a combination of elevations of the land (which are formed by peat hillocks and rollers, palsas, and hummocks) and its depressions, called hollows, which differ by the degree of watering and the

Table 1. Basic regional climatic characteristics of the tundra subzones in western Siberia: daily mean surface temperature (T), precipitation (P), sum of positive daily mean temperatures ($T_{>0}$), length of a period with a daily mean surface temperature of no less than 5°C (Veg), and area (S)

Tundra subzone	T , °C			P , mm/yr	$T_{>0}$, °C	Veg, day	S , thousand km ²
	year	January	July				
Arctic tundra	−10.9	−24.8	5.1 (August)	310	398	67	57.5
Typical tundra	−8.7	−24.3	9.8	288	827	96	110.0
Southern tundra	−8.2	−24.2	10.9	330	935	100	176.0

Basic climate characteristics are determined on the basis of data obtained at the Popov and Dickson meteorological stations (arctic tundra), Antipayupa and Marresale stations (typical tundra), and Mys Kamennyi and Novyi Port stations (southern tundra) from 1980 to 2005 (site of the All-Russian Research Institute of Hydrometeorological Information).

composition of vegetation. In such hollows there are inside wetland lakes that are small in area. Lakes in the tundra zone are subdivided according to the type of their origin into thermokarst, floodplain, and glacial. In the basins of drained thermokarst lakes called khasyryes, there are wetlands, palsas, and shallow lakes.

According to a digitized wetland typological map (Romanova et al., 1977), a significant area (about 58%) of the tundra zone in the region under consideration is weakly swamped-wetland ecosystems occupy less than 10%. On a small part (less than 9%) of the tundra territory in western Siberia, wetland landscapes occupy more than 80% of the area (Fig. 2). The most swamped territories are located in the northeast of the Yamal Peninsula, in the center of the Tazovskii Peninsula, and in the north of Gydanskii Peninsula.

Two key sites located in the typical and southern tundras of the Yamal–Nenets Autonomous Okrug (Fig. 1) were chosen for an analysis of the structure of landscapes in this region. Within these sites there are microlandscapes characteristic of the corresponding tundra subzones: palsas, peat hillocks and rollers, different hollows, small inside wetland lakes, lakes, and mineral soil ecosystems. In the summer of 2013, in these sites, field measurements were performed, during which meteorological parameters and the intensity of methane fluxes into the atmosphere from different tundra microlandscapes were measured, both vegetable and soil covers were studied, and the locations of the basic microlandscapes were determined.

According to the wetland typological map (Romanova et al., 1977), within the key typical-tundra site, there are both polygonal-roller and polygonal-fissure mires. Within the key southern tundra site, there are shrub-dominated tussock and flat-palsa bogs, whose peat hillocks are separated from one another by shallow mires. Khasyryes distributed throughout the key sites are partially wetland.

Lakes within the key sites are typical of northwestern Siberia: they are shallow (2–3.5 m in depth) and small (4×10^3 – 2×10^6 m² in area) water bodies with sloping shores that are wetland in places.

Images taken from onboard the Landsat 5 and 8 and WorldView-2 satellites were used in this work. Images taken from onboard the Landsat-5 and -8 satellites of medium-resolution within IR spectral ranges of 450–515 (blue), 525–605 (green), 630–690 (red), 750–900 (near-IR), and 1360–2350 (mid-IR) nm were used in our analysis. The spatial resolution of used spectral channels is 30 m per pixel, and the radiometric resolution is 8 bits. The images of these satellites cover an area of 36 200 km². The level of data processing includes radiometric correction, binding according to orbital parameters, and the standard UTM projection. Satellite (Landsat-5 and -8) data are free to access at <http://earthexplorer.usgs.gov>.

In addition to conventional spectral bands of 450–510 (blue), 510–580 (green), 630–690 (red), and

Table 2. Relative areas of wetland types in the tundra zone of western Siberia

Wetland types according to (Romanova et al., 1977)	Wetland area (% of the tundra-subzone area)		
	arctic tundra	typical tundra	southern tundra
Type 1	21.2	9.6	8.0
Type 2	7.0	9.4	2.7
Type 3	0.2	0.5	5.4
Type 4	0.0	0.0	0.1
Type 5	0.0	0.0	0.7

Polygonal-roller and polygonal-fissure mires (Type 1), polygonal mires combined with grass- and moss-dominated mires (Type 2), patterned (hollow and hollow-pool) flat-palsa bogs (Type 3), flat-palsa and high-palsa bogs (Type 4), and shrub-dominated tussock mires (Type 5).

770–790 (near-IR), the WorldView-2 satellite of ultrahigh resolution has four more spectral bands of 400–450 (violet), 580–620 (yellow), 700–740 (red-edge), and 860–1040 (near-IR 2). The spatial resolution of images amounts to 2 m for each channel. In this

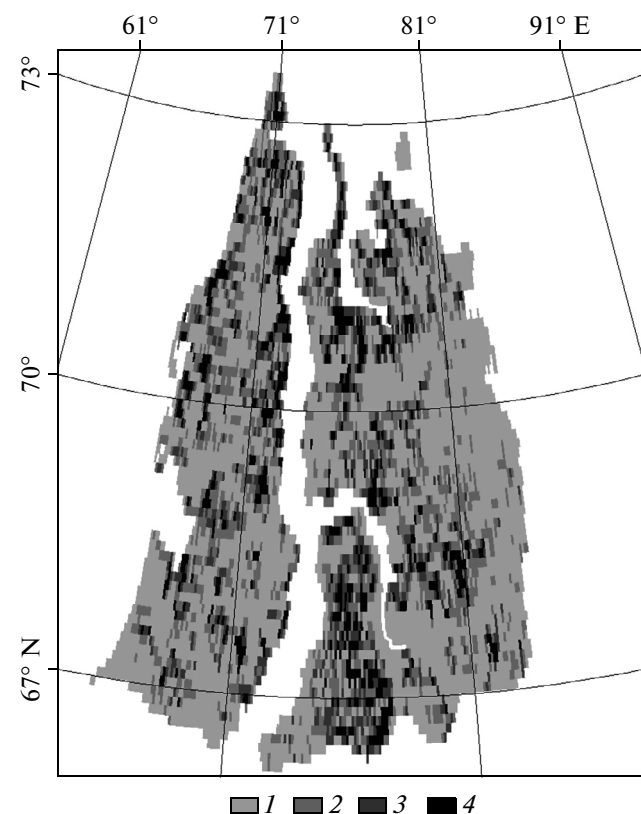


Fig. 2. Wetlands in the tundra zone of western Siberia: (1) less than 10%, (2) 10–50%, (3) 50–80%, and (4) over 80%.

Table 3. Area structure of tundra landscapes in western Siberia

Landscape type	Part of the landscape area in Landsat images, %		Part of the landscape area in WorldView-2 images, %	
	typical tundra	southern tundra	typical tundra	southern tundra
Water	8.0	7.6	14.6	4.8
Dry shrub-grass-lichen land on mineral soil	40.6	41.0	8.3	44.8
Dry land with shrubs on mineral soil	3.7	7.7	—	21.1
Khasyrey	13.1	14.8	11.3	3.8
Wetland khasyrey	8.0	4.8	4.2	4.7
Hollow (type 1)	11.2	10.1	8.3	3.7
The same (type 2)	8.8	1.6	30.3	1.8
The same (type 3)	—	0.5	—	2.0
Hillock-hollow complex	3.8	1.2	11.1	13.3
River floodplain	2.8	6.0	—	—
Sands, roads	0.0	3.6	—	—
Cloudy areas	—	1.1	—	—

study, satellite images in the UTM projection on the WGS-84 ellipsoid in the GeoTiff format with a radiometric resolution of 16 bits were used for the key southern and typical tundra sites 40 and 25 km² in area, respectively. The Obukhov Institute of Atmospheric Physics, Russian Academy of Sciences, has got the right to use data obtained aboard the WorldView-2 satellite.

The choice of satellite images was determined by the absence of snow cover in the key sites under consideration, the presence of minimum cloudiness in Landsat images, and the absence of cloudiness in WorldView-2 images. As a result, the satellite images of July 18, 2013 (Landsat-8), and September 13, 2011 (WorldView-2), were chosen to study the typical tundra key site. Satellite images of July 17, 2006 (Landsat-5), and June 22, 2012 (WorldView-2), were used to analyze the southern tundra key site. The Landsat images used in this study covered areas of 35300 and 29700 km² in typical and southern tundras, respectively.

The landscape structure of tundra ecosystems in western Siberia was analyzed on the basis of the satellite images and data obtained from field measurements. The images were deciphered and landscapes were classified using the GRASS 6.4.3 program package. In analyzing the Landsat images, classification with training was performed. Training polygons were chosen manually on the basis of expert evaluations and field-measurement data. About 30 training polygons were used for every class. The accuracy of classifications was determined using the kappa statistics method (Richards and Jia, 2006). In analyzing the WorldView-2 images, the method of principal components was used to reveal maximum differences in spectral radiance (Lillesand

et al., 2004). Classification with training was performed for the first three components. The classes singled out in the WorldView-2 images were compared with those in the corresponding Landsat images with the aid of the “r.statistics” module of the GRASS package.

RESULTS AND THEIR DISCUSSION

An analysis of the images of the Landsat satellite of medium resolution made it possible to differentiate 11 classes of tundra landscapes (Table 3). The accuracy of classification on the basis of the kappa-statistics method corresponds to a satisfactory level—the kappa indices amount to 85 and 72% for the images of southern and typical tundras, respectively. About 1% of the area in the image of southern tundra is covered by clouds, and it is impossible to identify landscapes on this part of the territory. According to results obtained from an analysis of satellite data, about 3 and 10% of the areas of typical and southern tundras, respectively, are occupied by floodplains, river sand drifts, and roads. Mineral soil ecosystems and dry khasyreys occupy about 60% of the areas of the tundra subzones under consideration. Wetland ecosystems and water bodies occupy approximately 38 and 28% of the areas of typical and southern tundras, respectively.

The tundra landscapes differentiated according to the Landsat images were analyzed using the more detailed images of the key sites taken aboard the WorldView-2 satellite. This analysis with the use of field-measurement data has made it possible to differentiate 19 classes of the land surface (microlandscapes) in southern tundra and 24 microlandscapes in typical tundra. Tables 4 and 5 give the results of classifying the microlandscapes of the key sites. The water

Table 4. Area structure of microlandscapes in the typical tundra of western Siberia

Microlandscape type	Part of microlandscapes area in satellite images, %	
	WorldView-2	Landsat
Water	11.8	12.6
Shallow water	3.0	1.9
Shallow water—floating mat	0.1	0.1
Dry land with shrub—lichen communities (type 1) on mineral soil	7.8	10.2
The same (type 2) on mineral soil	4.5	6.4
Dry land with lichen—shrub communities on mineral soil	5.4	4.9
Dry land with lichens on mineral soil	1.5	1.7
Lichens (type 1) on palsas and wetlands hummocks	3.1	1.0
The same (type 2)	4.8	1.5
Grass—shrub—moss communities in khasyrey	5.2	3.2
Grass—moss communities in khasyrey	1.9	1.2
Grass—green moss communities in khasyrey	1.5	1.1
Waterlogged sedge hollow (type 1) in khasyrey	6.0	5.8
The same (type 2)	0.5	0.3
Waterlogged sedge—sphagnum hollow (type 1) in khasyrey	1.5	1.8
The same (type 2)	0.3	0.4
Shrub—green moss communities on wetland polygons	9.2	10.8
Grass—moss communities on wetland polygons	6.3	4.4
Waterlogged cotton-grass-sphagnum hollow in the cracks on wetland polygons	5.4	4.7
Hollow with moss cover	1.9	1.4
Sedge—sphagnum hollow (type 1)	7.2	4.6
The same (type 2)	2.9	2.2
The same (type 3)	5.7	4.8
The same (type 4)	2.4	2.2

ecosystems of the key southern tundra site are subdivided into a deepwater part without water-surface vegetation (water microlandscape), a shallow-water part with sparse water-surface vegetation (shallow water microlandscape), and inside wetland lakes. The water ecosystems of the key site of typical tundra are represented by the following microlandscapes: water, shallow water, and “shallow water—floating mat.” The shallow water—floating mat landscape is a shallow plant-filled part of a water body. For the key sites of both typical and southern tundras, we have also differentiated microlandscapes for mineral soil ecosystems and dry khasyreys (7 and 5 types, respectively), wetland elevations (two types each), hollows (7 and 4 types, respectively), and waterlogged hollows (5 types each). It should be noted that the microlandscapes of the typical and southern tundras, which have the same name, differ in composition of vegetation and degree of watering.

In analyzing the WorldView-2 image of the key site in typical tundra, no microlandscape of inside wetland lakes was revealed. An additional visual expert analysis of this image allowed one to determine the location of sufficiently small (in area) water surfaces of inside wet-

land lakes and calculate their total area. The results of this analysis showed that, when classifying the microlandscapes of this satellite image, the inside wetland lakes were included in the classes of different waterlogged hollows. The results of calculating the area of the water surface of the inside wetland lakes showed that these lakes in typical tundra occupy less than 4% of the area of waterlogged hollows.

As an illustration, Fig. 3 shows the fragments of the deciphered Landsat and WorldView-2 images with the singled-out classes of landscapes and microlandscapes for the key site of the southern tundra. It follows from the fragment of the Landsat image that the key site under consideration involves all the basic classes of landscapes for southern tundra, which manifest themselves in this image. The fragments of the WorldView-2 image demonstrate to what extent the landscape structure of the territory under study may be more detailed if data are obtained from satellites of ultrahigh resolution when compared to data obtained from satellites of medium resolution: 8 and 10 microlandscapes are revealed in fragments (b) and (c) of the WorldView-2 image, respectively, while in the corresponding frag-

Table 5. Area structure of microlandscapes in the southern tundra of western Siberia

Microlandscape type	Part of microlandscapes area in satellite images, %	
	WorldView-2	Landsat
Water	7.1	7.3
Shallow water	3.8	5.2
Shallow lakes	2.7	4.9
Dry land with lichens on mineral soil	7.1	5.2
Dry land with lichen—shrub communities on mineral soil	4.2	3.5
Dry land with shrub—lichen communities on mineral soil	16.7	13.8
Dry land with low shrubs and lichens on hillocks	2.2	1.9
Dry land with shrubs on mineral soil	22.9	19.1
Shrubs on wetland polygons	3.9	3.0
Lichens on palsas and hummocks inside wetlands	2.7	2.2
Hollow with moss cover	1.1	0.9
Sedge—sphagnum hollow (type 2)	1.7	1.4
The same (type 2)	0.3	0.4
The same (type 3)	0.2	0.2
Waterlogged sedge hollow in khasyrey	0.2	0.2
Waterlogged sedge—cotton grass—sphagnum hollow	10.2	9.0
Waterlogged sedge—sphagnum hollow (type 1)	2.1	1.9
The same (type 2)	4.6	3.7
The same (type 3)	6.1	5.7

ments of the Landsat image only 5 types of landscapes are revealed for each fragment.

A comparison of the revealed classes of landscapes with the types of microlandscapes for the territories in the WorldView-2 images made it possible to determine the parts of the microlandscapes under consideration in each landscape class. On the basis of these data, the area was calculated for each microlandscape in the Landsat images under consideration (see Tables 4, 5). It should be noted that the parts of the microlandscape areas calculated for the WorldView-2 and Landsat images are sufficiently close to one another. According to satellite data, the wetlands in typical and southern tundras amount to 52 and 34%, respectively (Fig. 4). These wetland estimates exceed those obtained from calculations using a wetland typological map (23 and 32%, respectively) and only the Landsat images (32 and 18%, respectively). This fact implies that the area of wetland ecosystems is underestimated on the typological map of wetlands and that the accuracy of classifying landscapes on the basis of only Landsat images is lower. Ecosystems on mineral soils (dry land) are common in the tundra regions under study. According to obtained estimates, they occupy about 34 and 43% of the areas of typical and southern tundras, respectively. According to the typological map, the mineral soil ecosystems and lakes occupy almost 77 and 68% of the areas of typical and southern tundras, respectively.

Lakes in the regions under study occupy about 11 and 12% of the areas of typical and southern tundras,

respectively. According to estimates obtained for the area characteristics of microlandscapes in southern tundra, the area of the deep-water surface of lakes amounts to about 58.7% of their area and the area of the shallow-water surface of lakes amounts to 41.3%. The similar estimates for typical tundra amount to 81.7 and 17.5%, respectively. In the typical tundra region under study, a microlandscape of the littoral zone of lakes (shallow water—floating mat) was also revealed. The area of this microlandscape amounts to about 0.8% of the total area of lakes. It follows from these estimates that the tundra lakes have sufficiently large shallow-water zones ranging from 18 to 41% of the lake area. According to conclusions drawn by Bastviken et al. (2004), the littoral shallow-water zone of lakes occupies no more than 15% of their area.

Figure 5 gives the ratios between the areas of the basic groups of wetland microlandscapes for the tundra regions under consideration. An insignificant part (6–7%) of wetlands is occupied by hillocks and rollers, which form elevations in the landscapes of wetland ecosystems. Most wetland territories are occupied by hollows and small lakes. In the typical tundra region under study, hollow microlandscapes are most common and they occupy about 66% of the territory. In the southern tundra region under study, waterlogged hollows and inside wetland lakes occupy over 80% of the territory.

On the basis of the wetland typological map in western Siberia and data obtained from deciphering satellite images, the areas of the microlandscapes under consid-

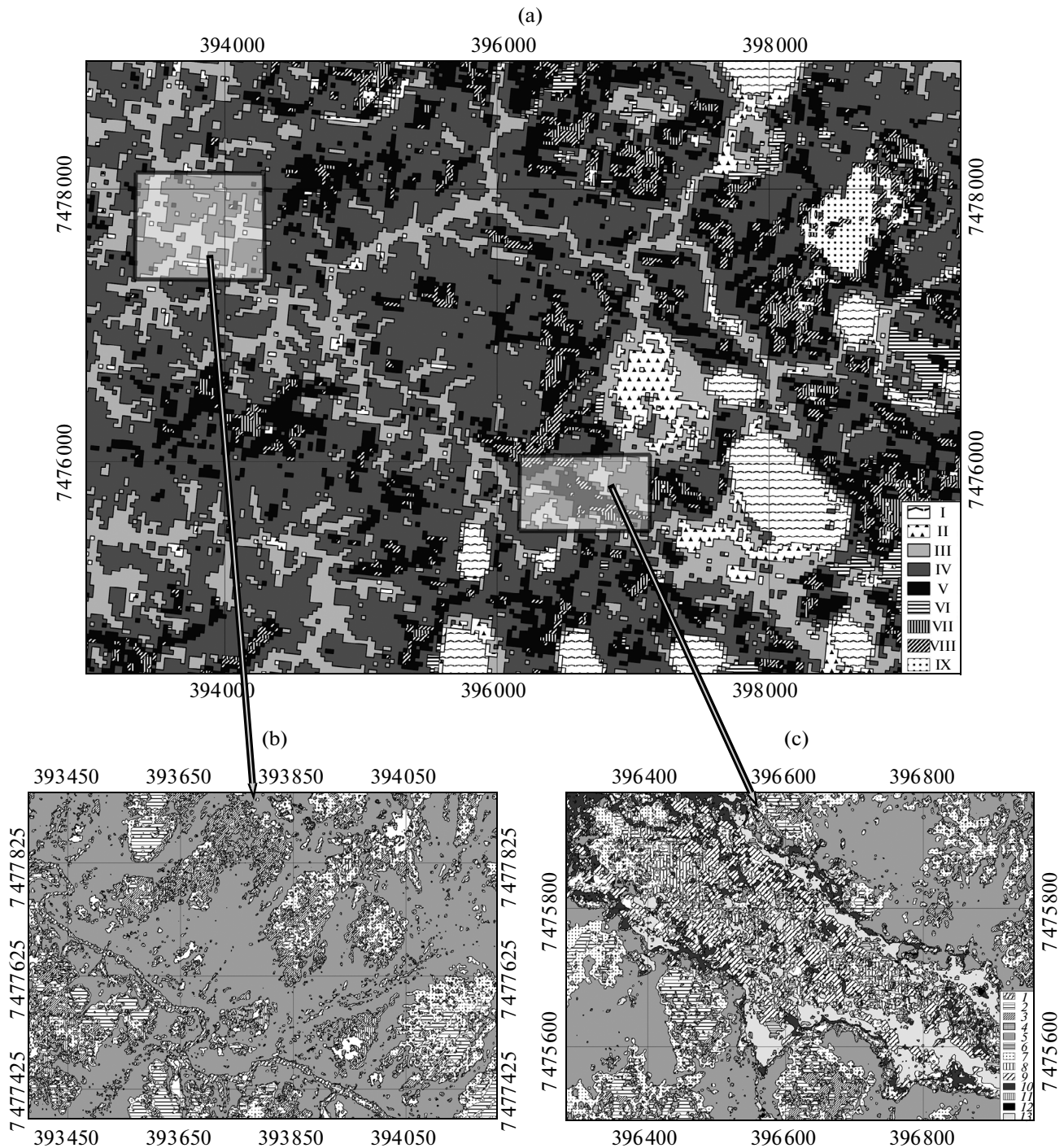


Fig. 3. Fragments of the deciphered (a) Landsat and (b, c) WorldView-2 images with the subdivided classes of landscapes and microlandscapes for the key site of southern tundra. Landscapes types: (I) water, (II) khasyrey, (III) dry land with shrubs on mineral soil, (IV) dry shrub-grass-lichen land on mineral soil, (V) hillock-hollow complexes, (VI) hollows (type 3), (VII) hollows (type 2), (VIII) hollows (type 1), and (IX) wetland khasyrey. Microlandscapes types: (1) shallow water, (2) dry land with lichens on mineral soil, (3) dry land with lichen-shrub communities on mineral soil, (4) dry land with low shrubs and lichens on hillocks, (5) dry land with shrubs on mineral soil, (6) waterlogged sedge-sphagnum hollow (type 1), (7) shrub-lichen communities on mineral soil, (8) shrubs on wetland polygons, (9) waterlogged sedge-sphagnum hollow (type 1), (10) waterlogged sedge-sphagnum hollow (type 3), (11) lichens on palsas and hummocks inside wetlands, (12) hollow with moss cover, and (13) waterlogged sedge-cotton grass-sphagnum hollow. The coordinates are given in the WGS84/UTM system.

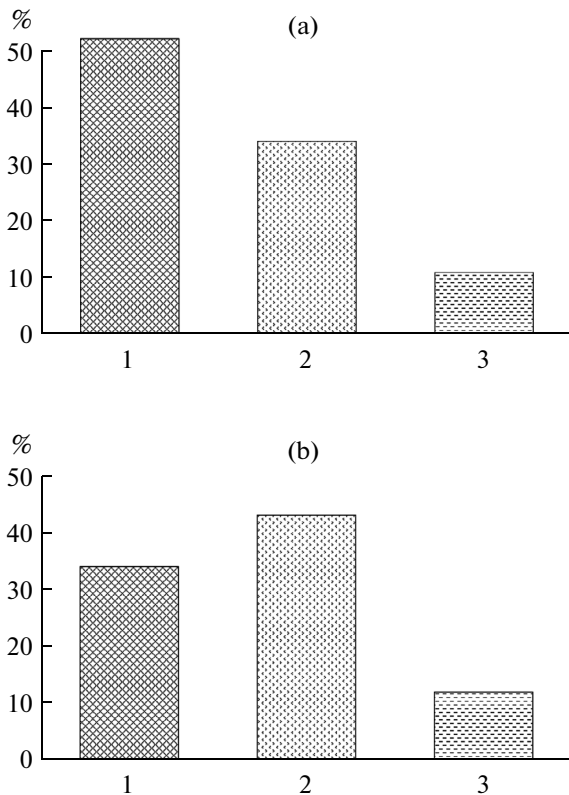


Fig. 4. Relative areas of (1) wetland and (2) mineral soil ecosystems and (3) lakes in (a) typical and (b) southern tundra subzones of western Siberia (% of the subzone area).

eration in different types of tundra wetlands have been estimated. Table 6 gives the calculation results. Since flat-palsa and high-palsa bogs ones, which occupy less than 0.1% of the area of southern tundra, are absent in the satellite images under analysis, the areas of microlandscapes were not determined for this type of wetlands and, therefore, they are absent in Table 6. It follows from the estimates obtained that hollow microlandscapes that differ in vegetation and watering level occupy over 78% of the area in all types of wetlands in both typical and southern tundras of western Siberia. The areas occupied by peat rollers and hillocks of different types in the wetland ecosystems for this region are insignificant (up to 8% of the area of bogs). The part of inside wetland lakes in typical tundra is small (up to 2%). The inside wetland lakes in southern tundra occupy from 9 to 14% of the wetland area.

CONCLUSIONS

The landscape structure of the tundra ecosystems of western Siberia has been analyzed on the basis of both Landsat and WorldView-2 satellite images and field-measurement data. For typical and southern tundras, 24 and 19 classes of microlandscapes, respectively, have been singled out. It should be noted that using the

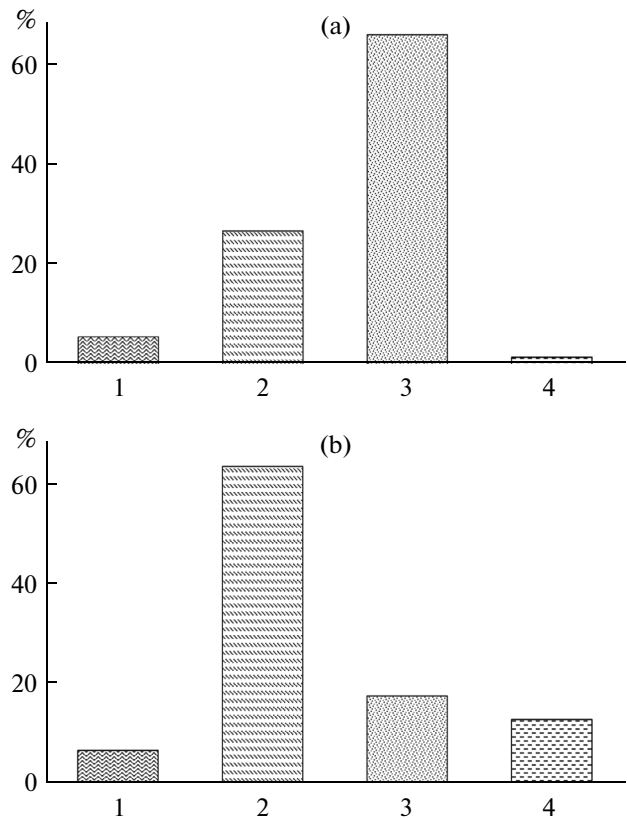


Fig. 5. Relative areas of wetland microlandscapes in (a) typical and (b) southern tundra subzones of western Siberia (% of the area of subzone bogs). Wetland microlandscapes: (1) hillocks and rollers, (2) waterlogged hollows, (3) hollows, and (4) inside wetland lakes.

results of ground-based measurements taken on the territories represented in satellite images significantly increases the accuracy in deciphering these images.

The results of this study showed that both wetland and lake ecosystems in the typical tundra of western Siberia occupy about 63% of the territory. The part of these ecosystems in southern tundra amounts to 46%. The estimates obtained for wetlands in the regions under study imply that such wetlands occupy somewhat larger areas when compared to those represented on the wetland typological map of western Siberia.

According to calculation results, a significant part (78–94%) of the wetlands area in the tundra zone of western Siberia is occupied by hollow microlandscapes. Hillocks and rollers in the wetlands of the region under consideration occupy 5–8% of the area. Shallow lakes are most common in the wetland ecosystems of southern tundra.

These results demonstrate that it is possible to simultaneously use images taken aboard satellites of both medium and ultrahigh resolutions in analyzing a complex mosaic structure of natural landscapes in large regions.

Table 6. Relative areas of wetland microlandscapes in the tundra zone of western Siberia

Wetland	Wetland microlandscapes area (% of the bog area)			
	hillocks, rollers	hollows	waterlogged hollows	inside wetland lakes
Typical tundra				
Type 1	6.3	65.1	26.9	1.7
Type 2	4.8	66.2	26.7	2.3
Type 3	5.2	65.8	28.3	0.7
Southern tundra				
Type 1	6.5	18.5	64.1	10.9
Type 2	7.1	18.1	65.4	9.4
Type 3	6.9	17.7	65.6	9.8
Type 5	7.5	20.2	57.9	14.4

The names of wetland types are given in Table 2.

The estimates obtained for the areas of wetland microlandscapes in the tundra zone may be used in solving problems related, for example, to simulating regional emissions of greenhouse gases from natural ecosystems on the basis of experimental data on these gases emitted into the atmosphere from individual microlandscapes or estimating the productivity of the vegetation of different wetlands on the basis of experimental data on the vegetation productivity of wetland microlandscapes.

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